

600 years. 8 systems. 1 map.

From Medici letters of credit to Europe's push for a digital euro — the complete settlement layer, explained.

WHAT'S INSIDE THIS DECK

01 Trust & Origins
Medici letters, Nostro/Vostro

02 SWIFT & Settlement
RTGS, DNS, CLS netting

03 Card Networks
\$25.8T — settles via Nostro

04 IPS & Real-Time
UPI, Pix, FedNow — 70+ countries

05 Cross-Border & QR
Nexus, DICT, bilateral links

06 Stablecoins & CBDC
\$33T, digital euro, sovereignty

\$7T+

Daily FX
Settlement

70+

IPS
Countries

130M

Wero+EuroPA
Users



33 slides. Zero fluff.

From Medici ledgers to digital euros. The complete settlement stack mapped.



The Trust Problem

How do you pay someone in a city where your bank doesn't exist?

BEFORE 1397: PHYSICAL SETTLEMENT



🏴‍☠️ Physical gold shipped by sea — weeks of transit

THE PROBLEMS



Robbery

Gold shipments raided en route



Currency

Florins ≠ English pounds



Speed

Weeks of transit by ship



No Trust

No shared institution



Physical gold moved for centuries.

International trade required physically moving value — until a Florentine banking family changed everything in 1397. → Next: how they did it.



The Medici Innovation

Letters of credit, 1397

THE FLOW



Medici Florence
Debit: Merchant A



Medici London
Credit: Merchant B

Letter of Credit = the message. No gold shipped.

THE CORE INSIGHT



**Don't move
the money**



**Move the
MESSAGE**



**Settle the
DIFFERENCE**



Unchanged for 600 years.

This principle powers every system in this deck. Only the speed of the message changes. → Next: the full timeline.



600 Years of Settlement

The message got faster.
The money still catches up.

SETTLEMENT SPEED BY ERA

YEAR	INNOVATION	SPEED
1397	Medici Letters of Credit	Weeks
1871	Telegraphic Transfers	Days
1973	SWIFT founded (15 countries)	1-5 days
1987	ISO 8583 — card protocol	T+2
2008	UK Faster Payments	Hours
2016	 UPI Launch (India)	<30s
2020	 Pix Launch (Brazil)	<10s
2024	 Wero Launch (Europe)	<10s
2026	 EU backs digital euro	Instant

Same principle, faster rails.



From weeks (ship) → days (telegraph) → seconds (IPS) → instant (stablecoins). The message always leads; settlement follows. But faster messages still need a place to settle. → Next: the account that makes it work.



Nostro / Vostro

The Medici innovation needed an account structure. This is it.

📌 Covered in yesterday's post

NOSTRO

"Our money at your bank"

DBS's view of its USD account held at JPMorgan in New York

VOSTRO

"Your money at our bank"

JPMorgan's view of that same DBS account sitting on JPM's books

~\$5T

Trapped
Liquidity

6,501

Active
Banks

-22%

Since
2011

↔ BIS CPMI 2022 quantitative review

Pre-funded = Capital trapped.



Banks must park trillions in accounts they don't control, across time zones they can't reach. This is why correspondent banking is expensive. → Next: follow \$50,000 through this system.

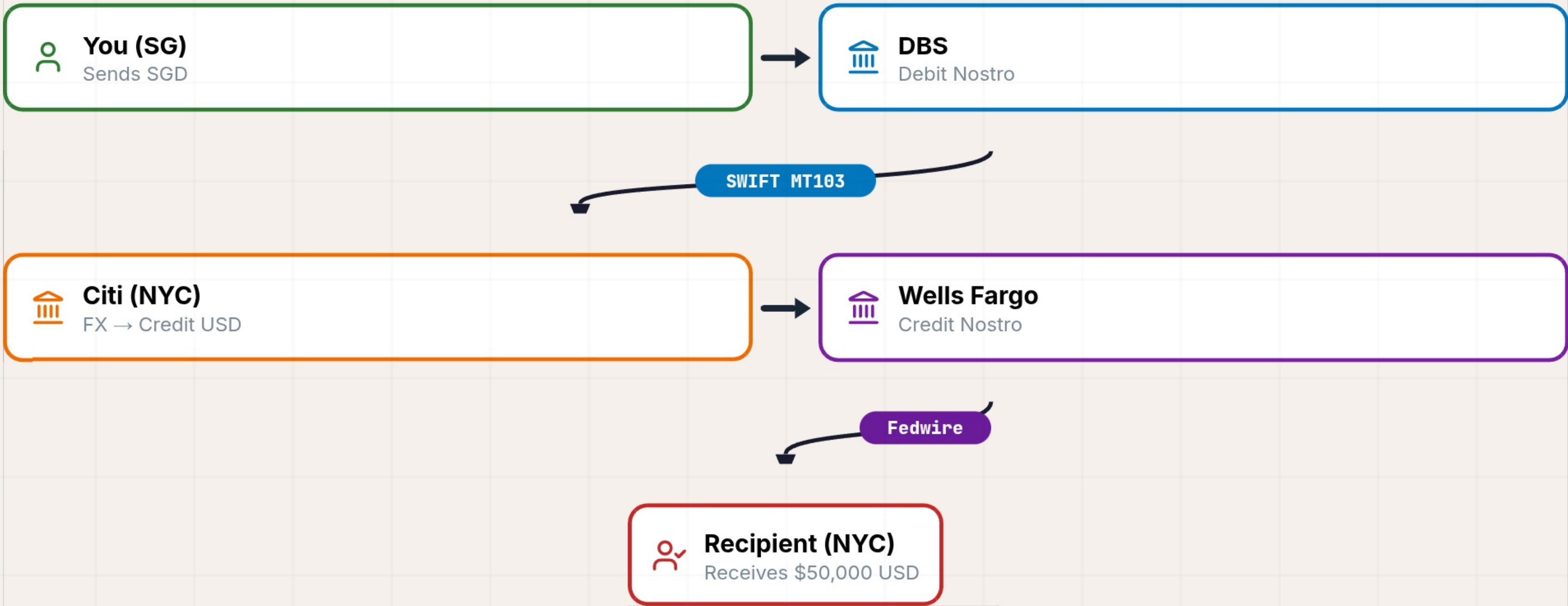


The \$50,000 Journey

4 banks. 48 hours. One payment.

Singapore → New York. Here's the path.

HERE'S HOW YOUR MONEY MOVES



24-96h

Transit
Time

\$25-65

Total
Fees

4

Ledger
Hops



Each hop = new ledger entry.

Money doesn't "move." Banks debit one account, credit another. → Next: SWIFT.



SWIFT

The \$50K journey used 4 banks. SWIFT is how they talked.



Doesn't move
MONEY



Moves
MESSAGES

11,500+

Institutions

53.3M

Msgs/Day

220+

Countries

MESSAGE EVOLUTION

MT → ISO 20022

MT103

999-char free text
Since 1977



pacs.008

Structured XML
Since Nov 2022



The wire doesn't carry cash.

It carries an instruction to move ledger entries at another institution. But what happens when the gross amounts are enormous? → Next: how banks avoid moving \$7T daily.



Multilateral Netting

Why D+2 is a feature, not a bug

WITHOUT NETTING

A→B: \$500

B→A: \$480

A→C: \$300

C→A: \$290

Total moved: \$1,570

▼ NET ▼

AFTER NETTING

A→B net: \$20

A→C net: \$10

\$30
total moved**98%**
reduction**\$7T+**

Daily CLS

>96%

Netted

18

Currencies

More txns = better netting. D+2 batching is a feature.



Without netting, banks would need 100× more liquidity. CLS settles both legs simultaneously (PvP). → But HOW do banks actually settle the net? Next: the two mechanisms.

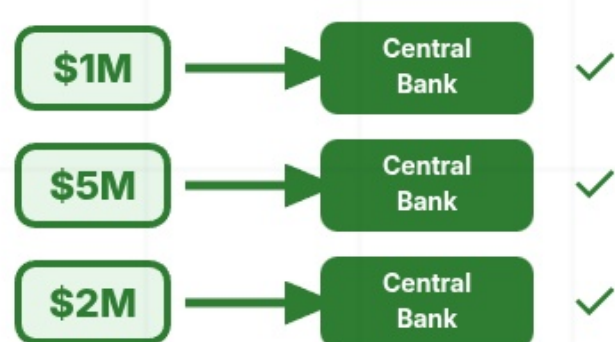


RTGS vs DNS

Two ways to settle. Same central bank.

RTGS

Real-Time Gross Settlement



Each txn settles individually

✗ Higher liquidity need

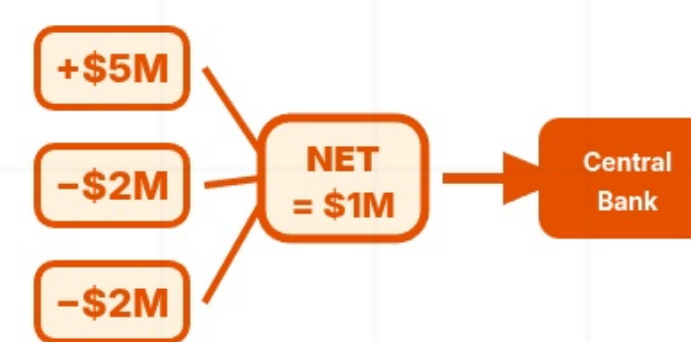
✓ **Zero counterparty risk**

Each payment is final the moment it clears the central bank ledger.

Pix • FedNow • TIPS

DNS

Deferred Net Settlement



Batched, net amount settles

✓ **Lower liquidity need**

✗ Some counterparty risk

Transactions accumulate, then one net position settles at end-of-cycle.

UPI • FPS • BACS

Where does IPS fit?

IPS (Instant Payment Systems) are consumer-facing products built **on top of** RTGS or DNS. Pix uses RTGS. UPI uses DNS. Both deliver instant payments to end users.

RTGS is the foundation. Everything else is a layer on top.



Whether gross or net, all roads lead to central bank settlement. → IPS is built on this. But so are cards. Next: how IPS actually works.

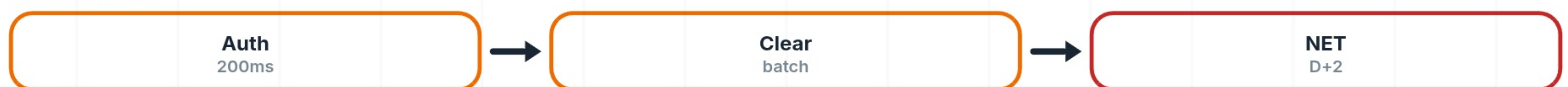


Instant Payment Systems

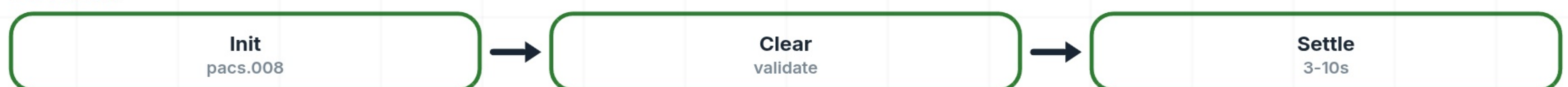
Not "like Venmo" — this is central bank infrastructure

CARD FLOW VS IPS FLOW

CARD FLOW:



IPS FLOW:



70+

Countries
(JPM report)

24/7

365 days
a year

Secs

Not days

🔗 JP Morgan Payments Sept 2025 report

IPS settles in central bank money.



Zero credit risk. No intermediary. The central bank is the ledger. → But cards still dominate POS. How do they actually settle? Next: the part most people miss.



Cards Settle Through Nostro/Vostro

The part most people miss

⚠ Visa/Mastercard are AUTHORIZATION networks. They don't have their own settlement rails.

HOW CARD SETTLEMENT ACTUALLY WORKS

1000 txns Bank A ↔ Bank B



Visa nets to 1 net amount



Net settles through CORRESPONDENT BANKING
Nostro/Vostro accounts

\$24T

gross/yr V+MC

99.7%

netted away

Netting is WHY T+2 exists.

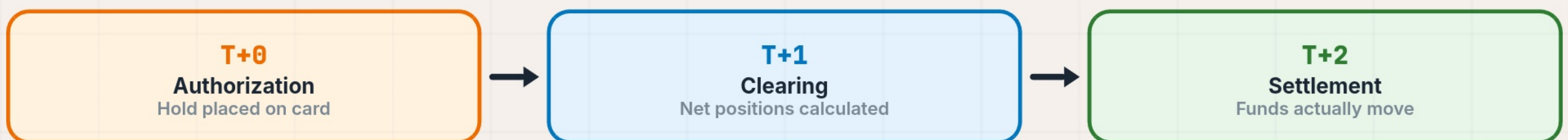


Without it, banks would need 100× more liquidity. The delay is the feature, not the bug. → Now that you know how settlement works, let's see cards in action. Next: the T+2 cycle.



The T+2 Cycle

Cards are the world's most popular payment method. But how do they actually settle?



DNS VS RTGS VS ACH

MECHANISM	HOW	SPEED	USED BY
DNS	Net, then settle batch	End of day	Card networks
RTGS	Settle each individually	Real-time	High-value
ACH	Batch low-value payments	2-3 days	Payroll, bills

\$794B

In-Flight
Daily

-22%

Cost Saving
(T+1)

2024

SEC moved
to T+1

DTCC, SEC Rule 15c6-1(a), BIS CPMI 2023

The "Approved" on your receipt is a promise.



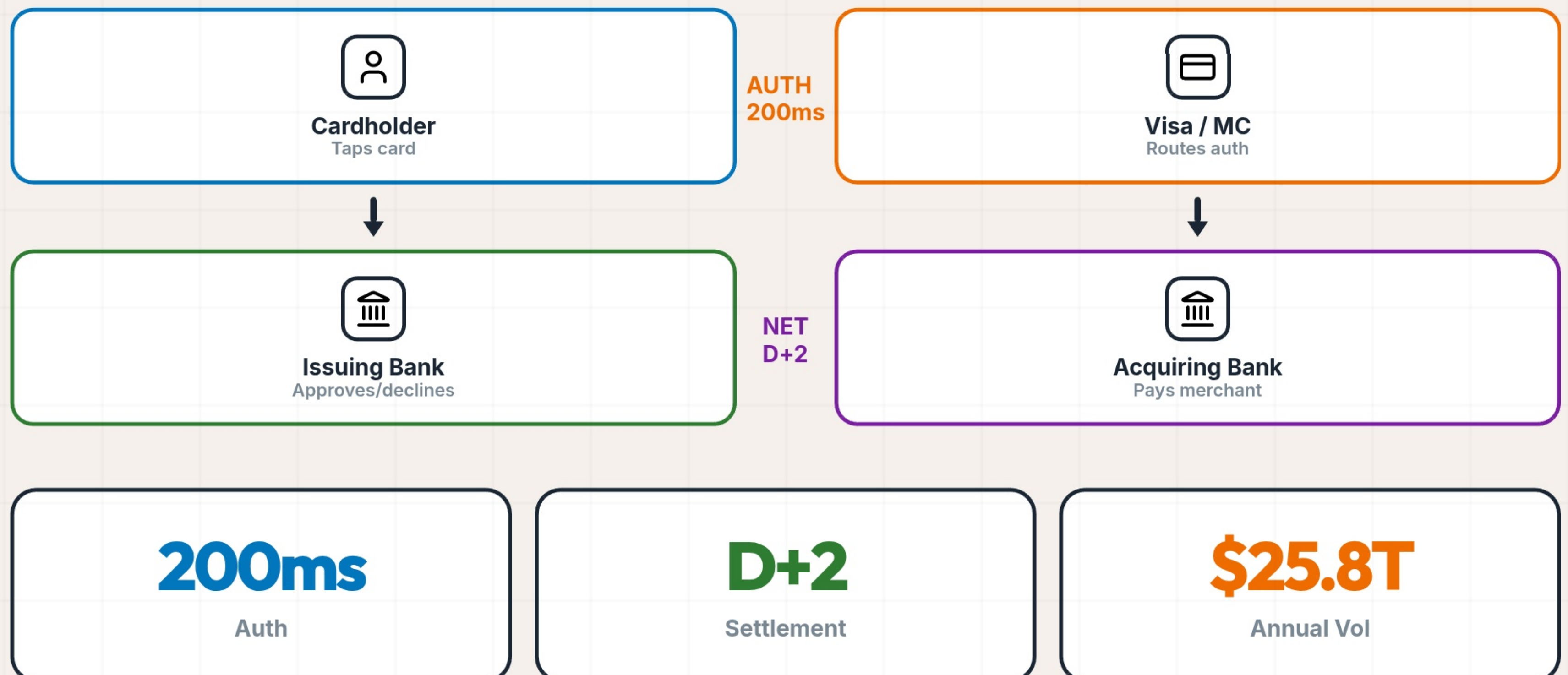
The merchant won't see real money for 2 business days. This gap is where risk, fees, and float live. → Auth happens in 200ms. But who processes it? Next: the 4-party model.



The 4-Party Model

Card networks authorize.
Banks settle.

☞ Scenario: You tap your credit card at a coffee shop for \$5. Here's what happens behind the scenes.



Cards don't settle. Banks do.



Visa/MC are authorization networks. Actual settlement? Via bank nostro accounts and central bank RTGS. Same plumbing since 1397. → Next: the binary protocol behind every tap.



ISO 8583

The binary protocol behind every card tap

📌 Deep-dived in: ISO 8583 carousel

MTI

4 digits

BITMAP

64 bits

DATA ELEMENTS

Up to 128 fields

~80B

ISO 8583

~2KB

JSON/REST

~5KB

ISO 20022

↔ Size comparison: 8583 vs REST vs 20022

500B+

Txns/Year

1987

First Spec

200ms

SLA

No JSON. No REST. Just bytes.



This 1987-era protocol processes \$25T/year. A bitmap tells the parser what's present — 25x smaller than REST. → But at 2-3% per transaction, countries are asking: is there a better way?



Card Settlement

How \$25.8T actually settles

THE NETTING CYCLE (DAILY)



FEE WATERFALL	
Interchange (issuer)	1.5-2.5%
Assessment (network)	0.13%
Acquirer markup	0.2-0.5%
Merchant pays total	~2-3%

150M+
Merchants

4.5B+
Cards

2-3% is the tax on digital commerce.

% D+2 settlement. ~3% fees. But: chargeback protection, 150M merchants, 200ms auth. → Every country with an IPS eventually asks: "Do we still need cards?" Next: the IPS revolution.



The IPS Revolution

Cards take D+2 to settle. IPS does it in seconds — in central bank money.

70+ COUNTRIES. ONE IDEA.



No intermediary
Direct bank-to-bank



Instant finality
Central bank money



Free (mostly)
0% consumer fee

THE IPS LANDSCAPE



UPI

16.6B txns/mo • 2016



Pix

313M txns/day record • 2020



PromptPay

74-79M txns/day • 2017



FedNow

1,000+ banks • 2023



The future is already here.

UPI processes more transactions monthly than Visa does globally. 70+ countries have IPS. But adoption varies wildly. → Next: the data.



The IPS Landscape

Not all "instant" settles the same way

SYSTEM	VOLUME	SETTLE
 UPI	21B/mo	DNS deferred net
 Pix	313M/day	RTGS instant gross
 FedNow	1,000+ banks	RTGS instant gross
 TIPS	€0.002/tx	RTGS via TARGET
 PayNow	via FAST	RTGS MEPS+
 PromptPay	74-79M/day	RTGS ITMX

🔗 Sources: NPCI, BCB, bot.or.th, ECB, Fed, MAS

UPI is the outlier.



India mandated zero MDR. Europe chose regulation. Different paths, same destination. → How is Europe doing it? Next: TIPS and the regulation lever.



TIPS

Europe's answer to UPI and Pix

📅 EU mandates SCT Inst acceptance – Oct 2025

HOW TIPS WORKS



€100K

Max per txn

<10s

End-to-end

24/7

365 days



Central Bank Money

Settles on ECB ledger, not commercial



ISO 20022 Native

pacs.008 structured end-to-end

The regulation is the adoption lever.



EUR instant was slow until the EU mandated it. Now every eurozone bank must accept SCT Inst by Oct 2025. → But what's the UX layer that makes IPS accessible to consumers? Next: QR codes.

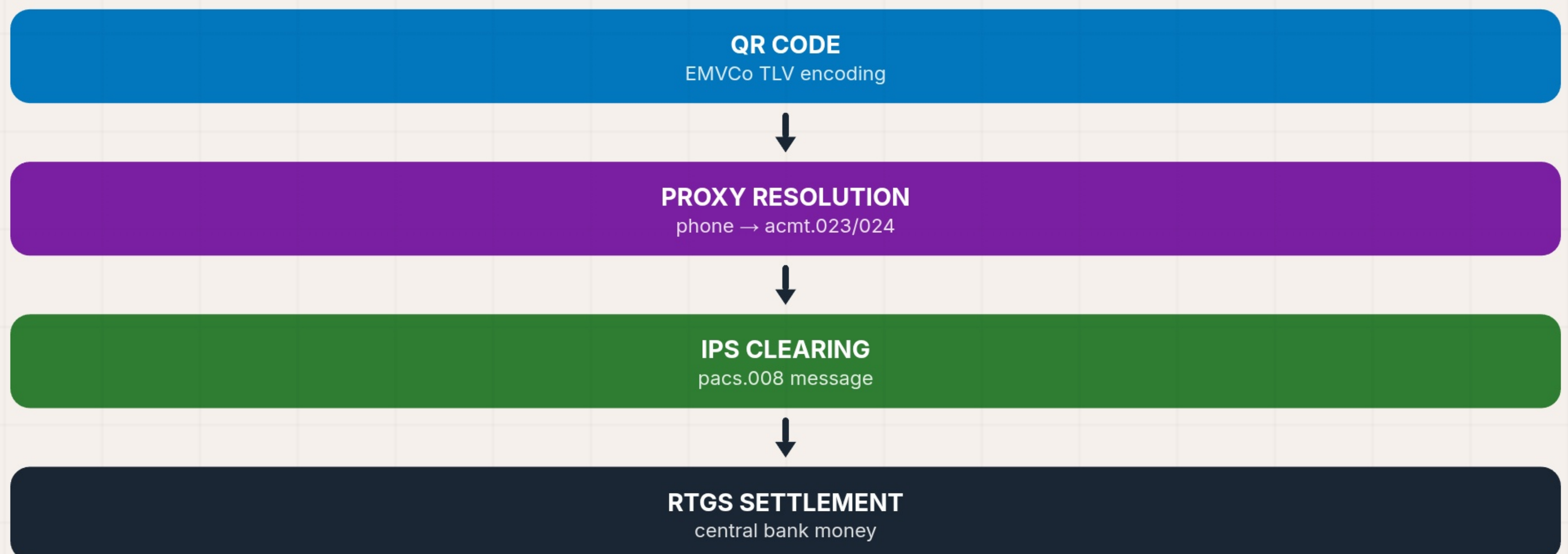


QR Payments

Not a payment system.
The UX LAYER on top of IPS.

📖 Our Feb 9 deep-dive showed the 7-layer stack behind every QR payment.

THE 4-LAYER QR STACK



QR just tells your phone WHERE to send the push.



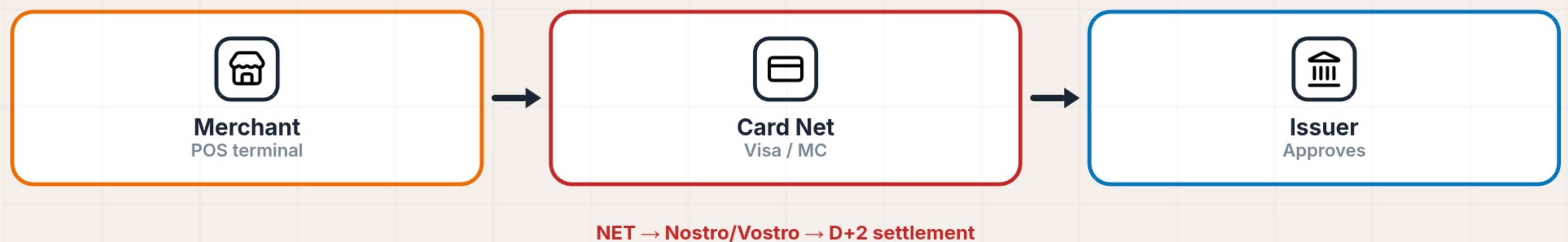
IPS does the settlement. The QR is an address, not a rail. → Same "Approved" message — completely different plumbing. Next: cards vs IPS, side by side.



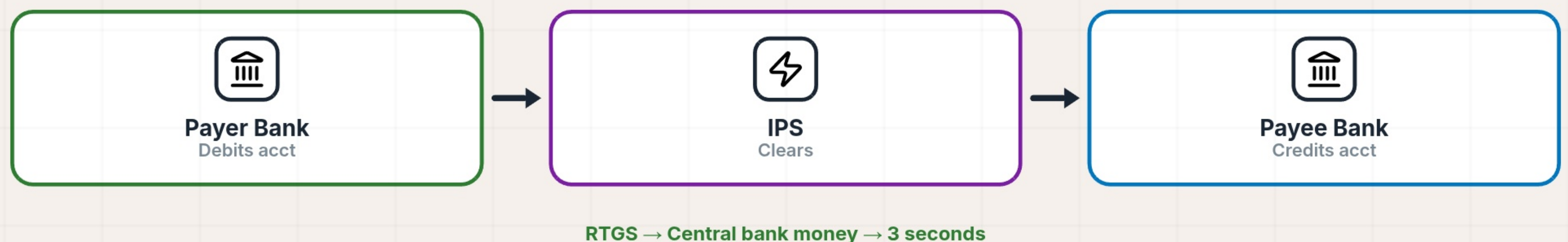
QR vs Card Flow

Two models. Same goal. Very different paths.

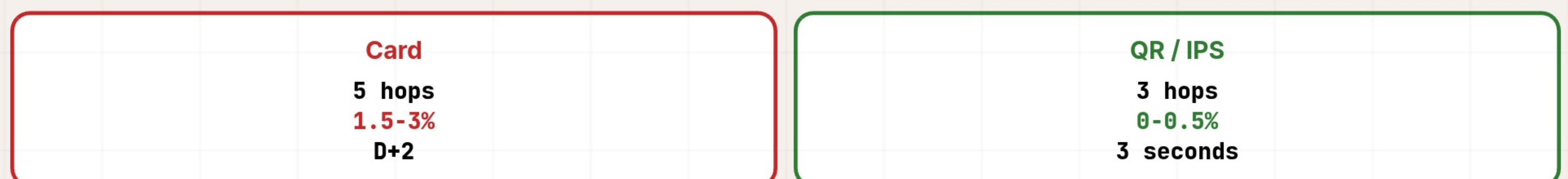
CARD (PULL MODEL)



QR (PUSH MODEL)



HEAD TO HEAD



Same outcome. Different architecture.



Cards win on chargebacks and 150M merchants. IPS wins on speed and cost. Both coexist. → Next: sovereignty.



Wero

Europe's home-grown alternative to Visa and Mastercard

📄 EPI + EuroPA sign MoU – Feb 2026

WHAT IS WERO?

Built on SEPA Instant (SCT Inst) — TIPS rail
Users 130M (16 countries by 2027)
Phase P2P live → POS 2025 → e-commerce 2026
Owners 16 EU banks (EPI consortium)

WHY THIS MATTERS



Visa + Mastercard

56-57% of EU
non-cash payments

US-controlled infrastructure
2-3% interchange fees



Wero

Account-to-account
0% interchange

EU-owned by 16 banks
Built on SEPA Instant

Payment sovereignty isn't abstract.



When 56% of your digital payments flow through foreign infrastructure, you're one geopolitical event away from disruption. → If IPS works domestically, can it work across borders? Next: the hardest problem in payments.



Cross-Border IPS

IPS solved domestic payments. Now the hard part: crossing borders.

WHY IT'S HARD



Time Zones

SG is +8, NYC is -5. Settlement windows don't overlap.



FX Complexity

Real-time rates, spread, who bears the risk?



AML/KYC

Different regulations per jurisdiction



Sovereignty

Central banks won't share RTGS access

THREE APPROACHES

Hub Model

BIS Nexus
Multilateral

Bilateral Links

UPI ↔ PayNow
1-to-1

Stablecoins

USDC corridors
Gap-filler

Domestic IPS solved domestic payments.



But a Thai tourist in Singapore still can't use PromptPay to pay a NETS merchant. Cross-border is the frontier. → Bilateral links don't scale: N countries = N^2 links. Next: the hub model.



Nexus

One hub. Many IPS. Instant cross-border.

Our Feb 5 carousel covered Nexus in depth.

TODAY: BILATERAL MESH (N^2 LINKS)



NEXUS: STAR TOPOLOGY ($N \times 1$ LINKS)



$N \times 1$
connections

$\leq 60s$
target SLA

2026
go-live

Nexus solves the coordination problem.

Connect once, reach all. FXPs bid competitively. → But are G20 targets being met? Next: the gap.



Local Card Networks

Domestic sovereignty, lower fees

DOMESTIC ALTERNATIVES TO VISA/MC

**RuPay**

60%+ India domestic share

**Elo**

25% Brazil domestic

**UnionPay**

90%+ China domestic

**Mir**

Post-sanctions domestic

**mada**

Saudi Arabia — mandatory

**JCB**

Japan — 140M cards

Russia's lesson.



When Visa/MC exited Russia in 2022, domestic payments were disrupted overnight. Every major economy now maintains a domestic card rail. → But global remittance costs are still \$12.70 per \$200. Who fills the gap? Next: the G20 scorecard.



G20 Roadmap

Cross-border targets by 2027

THE 4 TARGETS

\$3Max cost per
\$200 remittance**1hr**75% of payments
credited within**100%**Payment corridors
at <5% cost**24/7**Access in every
corridor

GAP ANALYSIS: TARGET VS REALITY

\$3

Target

vs

\$12.70

Current avg

24% there

FSB Cross-Border Payments Progress Report 2024

The gap is still massive.



\$12.70 per \$200 remittance vs the \$3 target. 6,501 correspondents — down 12% since 2011. Corridors are going dark. → Stablecoins entered this gap. Next: what they actually solve — and what they don't.



Stablecoins

Correspondent banking is slow. IPS is domestic. What fills the gap?

\$33T

Annual Transfer
Volume (2024)

\$230B+

Market
Cap

WHERE STABLECOINS WIN



De-risked corridors

Nigeria, Turkey, Argentina — places correspondent banks avoid



24/7 settlement

No banking hours, no weekends



Low cost

<\$0.01 on Solana/Base



Programmable

Smart contract escrow, conditionals

WHERE THEY DON'T

Not legal tender

Counterparty risk

Reserve opacity

On-chain in 15 seconds. But...



100% pre-funding. No credit extension. USDC de-pegged to \$0.88 during SVB. Off-ramp to fiat adds 1-3 business days. → But where correspondents won't go, stablecoins fill the gap. Next: who's moving real volume.



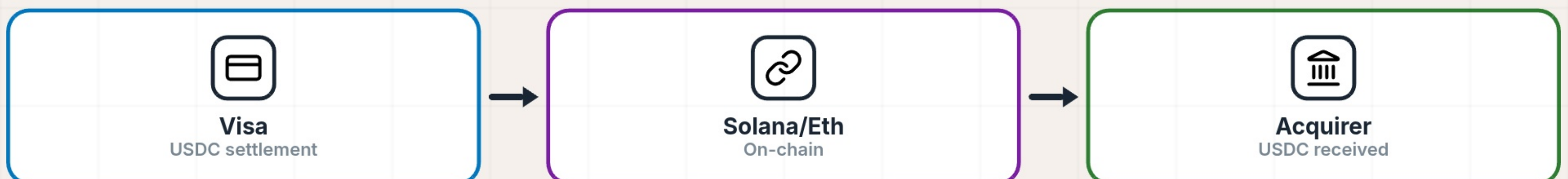
Institutional Stablecoins

When TradFi meets crypto rails

MAJOR PLAYERS

USDT	Tether — largest by cap	\$143B
USDC	Circle — regulated, transparent	\$58B
JPM Coin	JPMorgan Kinexys	\$5B/day

VISA'S STABLECOIN PLAY



↔ Visa expanded USDC settlement to Solana (Sep 2023)

\$5B

Kinexys
Daily Vol

24/7

Settlement
Window

JPMorgan moves \$5B/day on-chain.



Same institutions. New rails. Kinexys isn't a crypto experiment — it's production infrastructure. Visa settles USDC on Solana. → But stablecoins still need fiat off-ramps. What if the central bank issued the token? Next: CBDCs.



CBDCs

Stablecoins are private money. But what if the central bank went digital?

WHAT MAKES A CBDC DIFFERENT



Central bank
issued



Legal
tender



Offline
capable

134

Countries
Exploring

66

Active
Projects

3

Fully
Launched

↗ Atlantic Council CBDC Tracker, 2024

RETAIL VS WHOLESALE

Retail

Public use. Replaces cash. Wallet-based. Digital Euro.

Wholesale

Interbank. FX settlement. mBridge (BIS).

CBDCs are digital cash, not digital banking.



A CBDC is a central bank liability. Your bank deposit is a commercial bank liability. Different risk profiles. Like stablecoins, CBDCs are another option — not a replacement. → Next: the regulation framework.



MiCA

Europe regulates crypto-assets.
The template for the world.

📅 MiCA fully enforced – Jun 2024

WHAT MiCA REQUIRES



Reserve backing

1:1 reserves, audited, segregated



White paper

Mandatory disclosure to users



Licensing

All CASPs must be authorized



Volume caps

€200M daily for non-EUR stables

IMPACT ON STABLECOINS



USDT delisted

From EU exchanges



USDC compliant

Circle got license

MiCA is the template.



UK, Singapore, Hong Kong are all modeling their stablecoin regulations on MiCA's framework. First-mover advantage in regulation. → But Europe isn't just regulating. They're building. Next: the digital euro.



Digital Euro

CBDCs are theory. Europe is building one. Here's why.

📅 Feb 2026: EU Parliament backs online + offline CBDC

"Every time a European taps a card, the transaction flows through United States infrastructure."

– Christine Lagarde, ECB President

THE RUSSIA LESSON

2022



Visa/MC cut Russia

Cards stopped working overnight



Domestic payments disrupted

Foreign dependency exposed

Europe asked: "What if that's us?"

56-57%

EU Card
Dependence

\$24T

V+MC
Annual

~2029

Launch
Estimate



This isn't about technology. It's about sovereignty.

Europe's strategic response to US payment infrastructure dominance. → Next: the complete map.



The Settlement Stack

Five models. Same goal. Different tradeoffs.

MODEL	 Correspondent Banking	 Instant Payments	 Cross-Border Hubs	 Stablecoins	 CBDC
 Speed	24-96h	3-10s	≤60s	~15s	Instant
 Cost	\$25-65	\$0-0.50	~\$1	<\$0.10	\$0
 Settles	RTGS (T+2)	RTGS or DNS	Central Bank	On-Chain	Central Bank
 Reach	Global	Domestic	X-Border	Global	Sovereign
 Example	SWIFT Fedwire	UPI, Pix FedNow	Nexus  Future	USDC USDT	e-CNY mBridge



Not competitors. Interdependent layers.

IPS settles through RTGS. Stablecoins bridge gaps. Nexus connects them cross-border. CBDCs could unify the base layer. → What stays constant? Next: the 5 principles.



5 Principles That Never Changed

600 years. Same rules.

1

Move the message, not the money

Medici letters → SWIFT MT103 → blockchain txs. The instruction travels; ledgers reconcile.

2

Net everything, settle the difference

CLS nets \$7T → \$700B daily. Visa nets billions of transactions into one settlement.

3

Central bank money = finality

TIPS, FedNow, TARGET2 all settle in central bank accounts. No counterparty risk.

4

Sovereign infrastructure = control

Every country building IPS owns its payment rails. India's UPI, Brazil's Pix — sovereign by design.

5

The message gets faster; settlement follows

Weeks (ship) → days (telegraph) → seconds (IPS) → instant (stablecoins). Same arc, forever.



Technology changes. Money principles don't.

Whether it's a Medici letter, a SWIFT MT103, or a stablecoin — the same 5 rules govern settlement.



The Settlement Layer.

600 years of evolution in 33 slides.

THE LEGACY

Correspondent Banking

Medici letters → SWIFT MT103. Credit-based, \$7T+/day. Still the backbone.

THE REVOLUTION

Real-Time IPS

UPI 228B, PIX \$5T, FedNow. Push payments settle in seconds, not days.

THE DISRUPTOR

Stablecoins

\$33T/yr volume. 24/7, atomic settlement. MiCA regulates, USDC complies.

THE MULTIPLIER

CBDCs + Nexus

134 countries exploring. Nexus connects 60+ IPS. Sovereign digital money.

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Sivasubramanian R.

Product Owner • Payments & AI



Technology changes. Money principles don't.

Whether it's a Medici letter, a SWIFT MT103, or a stablecoin — the same 5 rules govern settlement. Which layer is your market on?



Save

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Comment

What layer are you on?